

Day_5_Visulization with seaborn

September 11, 2026

0.1 Day 5 - Visualization with Seborn

0.2 Objective

The objective of this task is to visualize the customer churn dataset using seaborn.

Different Visulization techniques such as Distribution Plot, Count Plot,Heatmap, and Pair Plot are used to understand data distribution, category counts, correlations, and relationships between numerical variables.

```
[4]: import pandas as pd

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)

df
```

```
[4]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	
17	1018	39	32	80	One year	
18	1019	53	54	100	Two year	

19 1020 33 16 70 Month-to-month

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

0.3 1. Distribution Plot

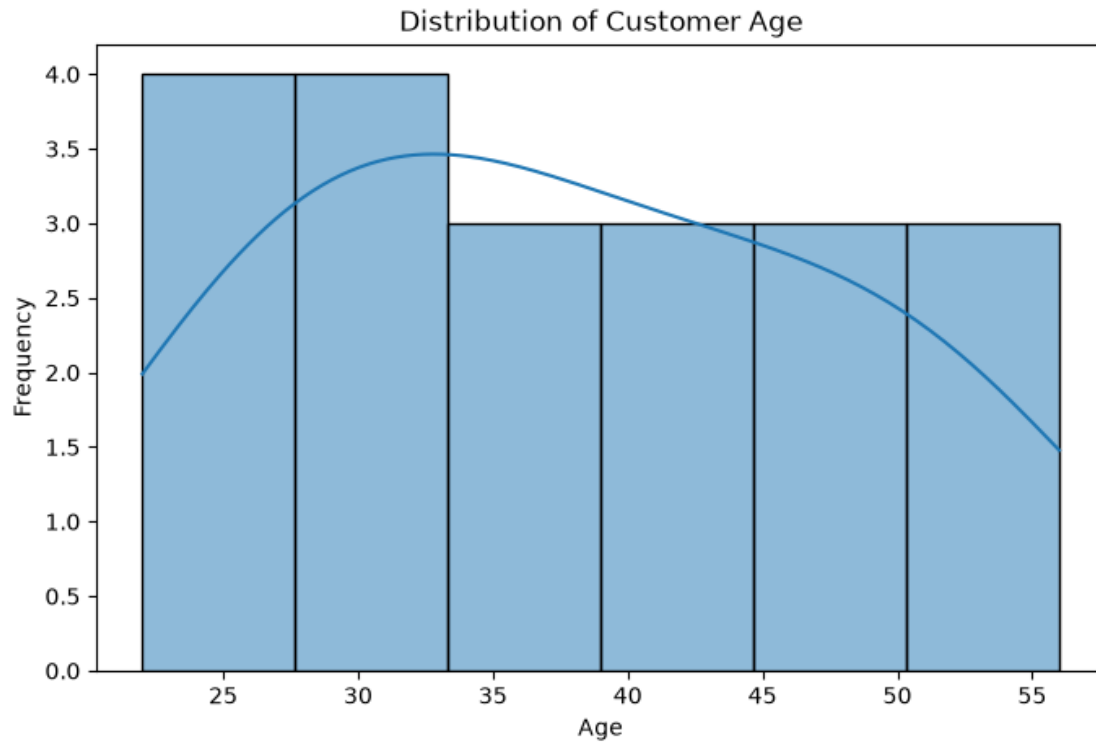
A distribution Plot is used to understand how numerical data is distributed.

In this task,the distribution of customer Age is visualized using a histogram with a density curve.

The `displot()` function is deprecated in newer versions of Seaborn, so `histplot()` is used as the modern alternative.

```
[5]: plt.figure(figsize=(8,5))
sns.histplot(df["Age"], kde=True)
plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



0.4 2. Count Plot

A Count Plot Is used to display the number of observations in each categorical group.

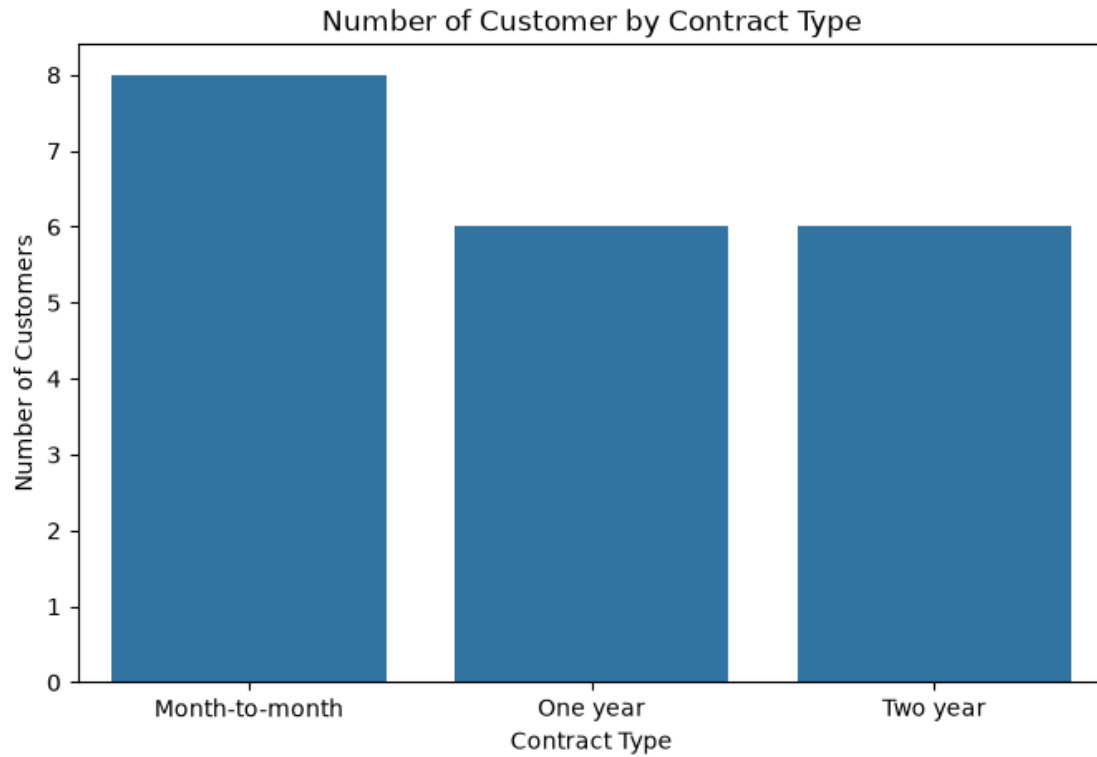
In this task, the number of customers in each Contegory is visualized.

```
[7]: plt.figure(figsize=(8,5))

sns.countplot(x="Contract", data=df)

plt.title("Number of Customer by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")

plt.show()
```



0.5 3. Heatmap

A heatmap is used to visualize relationships or correlations between numerical variables using a colors.

In this task, a correlation heatmap is created for the numerical columns of the Customer Churn Dataset.

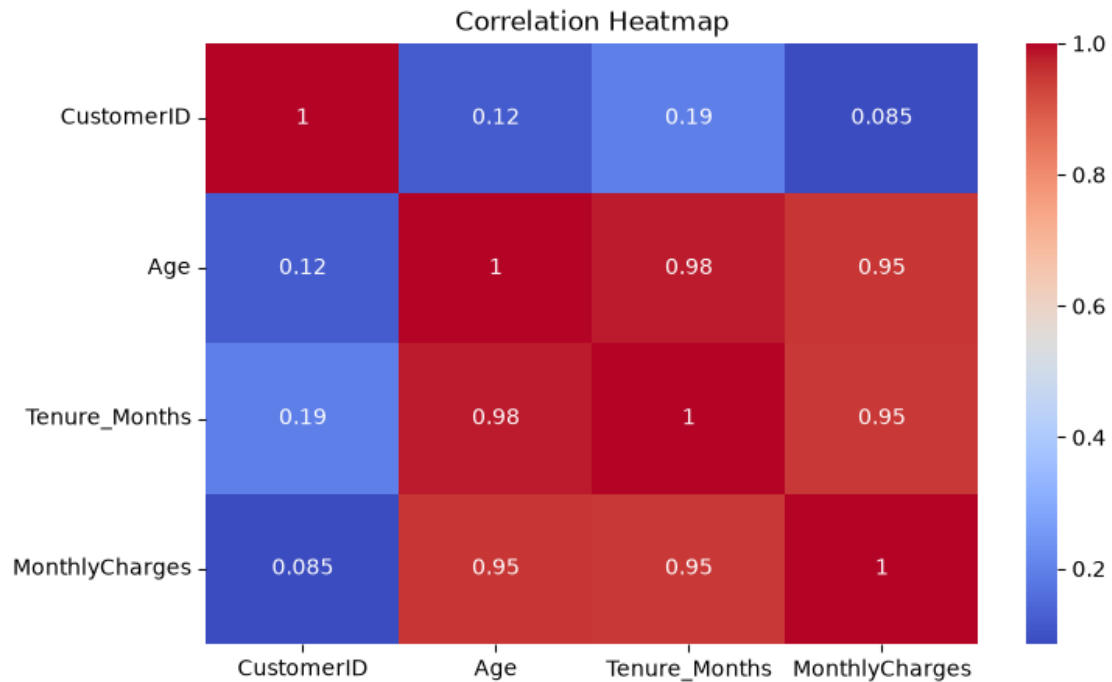
```
[8]: plt.figure(figsize=(8,5))

correlation=df.corr(numeric_only=True)

sns.heatmap(correlation, annot=True,cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()
```

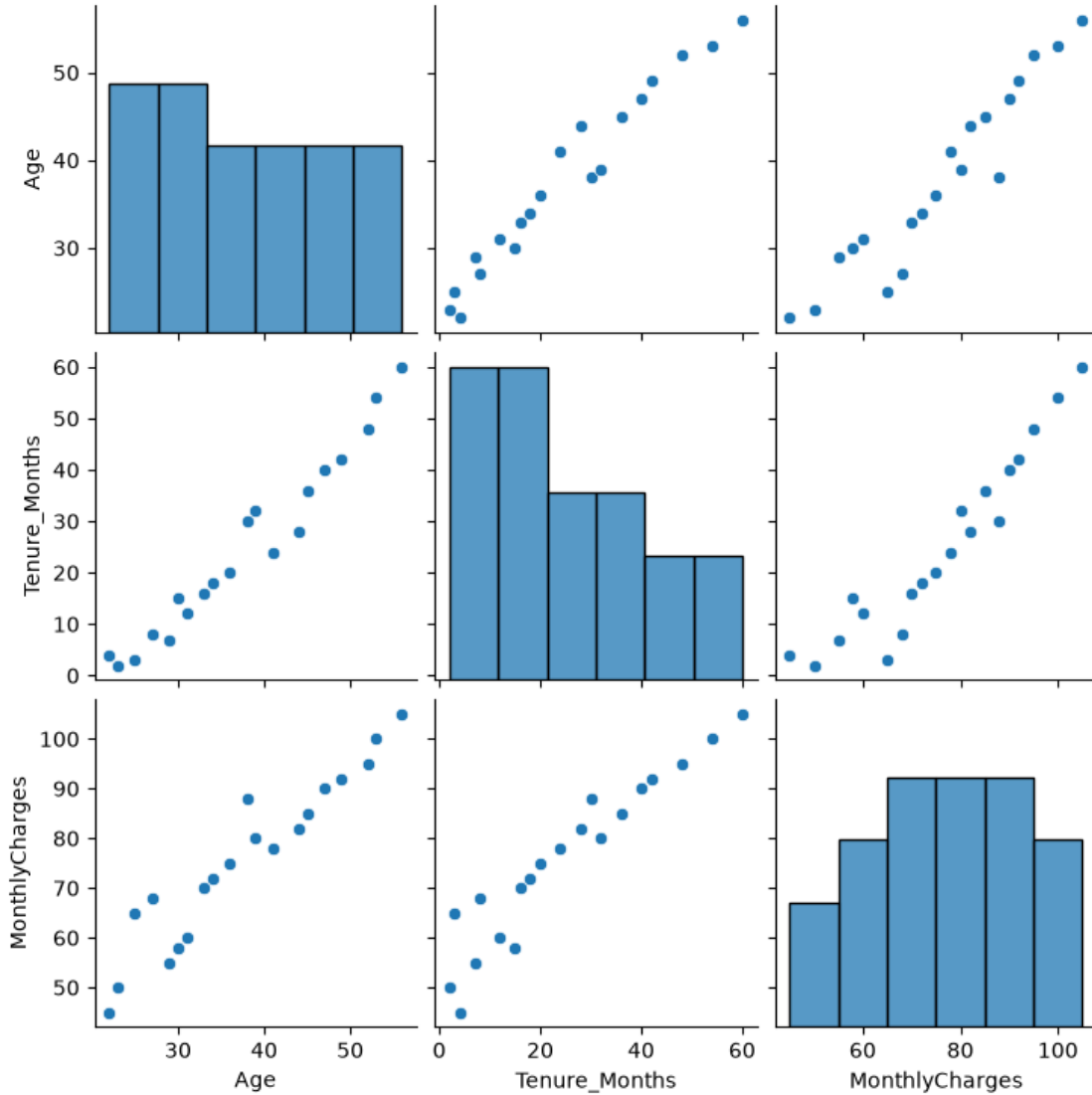


0.6 4. Pair Plot

A Pair Plot is used to visualize pairwise relationships between multiple numerical variables.

In this task, Age, Tenure_Months, and MonthlyCharges are compared to understand their relationships and distributions.

```
[9]: sns.pairplot(  
      df[["Age", "Tenure_Months", "MonthlyCharges"]]  
    )  
  
plt.show()
```



0.7 Conclusion

In this task, different visualization techniques were performed using Seaborn.

The Distribution Plot was used to understand the distribution of numerical data, the Count Plot was used to compare categorical values, the Heatmap was used to visualize correlations between numerical variables, and the Pair Plot was used to understand relationships between multiple numerical variables.

These visualizations provide better insights into the dataset and help in further data analysis and machine learning tasks.